WORKSHEET 7.6

Particular Solutions and Their Domains

Unit 7: Differential Equations

Key ideas

- Use the initial condition to find the constant **before** solving for y where possible; the algebra is easier.
- When a square root appears, the initial condition decides the sign.
- When an absolute value appears, the sign of the quantity at the initial point decides how to remove it.
- The **domain** of a particular solution is the largest single interval, containing the initial x-value, on which the solution is defined and satisfies the equation.

Worked examples

Example 1 · An exponential

Solve $\dfrac{dy}{dx} = \dfrac{y}{3}$ with $y(0) = 6$.

Solution. $\ln|y| = \dfrac{x}{3} + C$, so $y = Ae^{x/3}$ with $A = 6$.

$y = 6e^{x/3}$.

Example 2 · Choosing a sign

Solve $\dfrac{dy}{dx} = \dfrac{4x^3}{y}$ with $y(0) = -2$.

Solution. $\dfrac{y^2}{2} = x^4 + C$ with $C = 2$, so $y^2 = 2x^4 + 4$. The initial value is negative.

$y = -\sqrt{2x^4 + 4}$.

Example 3 · A domain

Solve $\dfrac{dy}{dx} = \dfrac{y^2}{x^2}$ with $y(1) = \dfrac{1}{2}$, and state the domain.

Solution. $-\dfrac{1}{y} = -\dfrac{1}{x} + C$; at $\left(1, \frac{1}{2}\right)$: $-2 = -1 + C$, so $C = -1$.

$y = \dfrac{x}{x+1}$, **for** $x > 0$ **(the equation is undefined at** $x = 0$**).**

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Practice

CORE

1. Solve $\frac{dy}{dx} = -4y$ with $y(0) = 3$.

2. Solve $\frac{dy}{dx} = 2x$ with $y(1) = 5$.

3. Solve $\frac{dy}{dx} = \cos x$ with $y(0) = -1$.

4. Solve $\frac{dy}{dx} = \frac{1}{x}$ for $x > 0$ with $y(1) = 2$.

EXAM LEVEL

5. Solve $\frac{dy}{dx} = \frac{9x}{y}$ with $y(0) = -3$.

6. Solve $\frac{dy}{dx} = 3(y - 2)$ with $y(0) = 5$.

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7. Solve $\dfrac{dy}{dx} = e^y$ with $y(0) = 0$, and state the domain of the solution.

8. Solve $\dfrac{dy}{dx} = y\cos x$ with $y(0) = 4$.

Question 9 · Free-response practice

Consider the differential equation $\dfrac{dy}{dx} = x(y-1)^2$. Let $y = f(x)$ be the particular solution with $f(0) = 3$.

(a) Find the slope of the graph of f at $(0, 3)$. What does this tell you about the graph there?

(b) Find the particular solution $y = f(x)$.

(c) State the domain of the solution, and give a reason.

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Worked solutions

1. $y = 3e^{-4x}$.
2. $y = x^2 + C$ with $C = 4$: $y = x^2 + 4$.
3. $y = \sin x - 1$.
4. $y = \ln x + 2$.
5. $y^2 = 9x^2 + C$ with $C = 9$; the initial value is negative: $y = -3\sqrt{x^2 + 1}$.
6. $\ln|y - 2| = 3x + C$; $y - 2 = 3$ at the start: $y = 2 + 3e^{3x}$.
7. $-e^{-y} = x + C$ with $C = -1$: $e^{-y} = 1 - x$, so $y = -\ln(1 - x)$, for $x < 1$.
8. $\ln|y| = \sin x + C$: $y = 4e^{\sin x}$.
9. (a) $0 \cdot (2)^2 = 0$: the tangent is horizontal at $(0, 3)$.

   (b) $\dfrac{dy}{(y-1)^2} = x\,dx$ gives $-\dfrac{1}{y-1} = \dfrac{x^2}{2} + C$, and $C = -\dfrac{1}{2}$. So $y - 1 = \dfrac{2}{1 - x^2}$, that is $y = 1 + \dfrac{2}{1 - x^2}$.

   (c) $-1 < x < 1$: the solution is undefined at $x = \pm 1$, and the domain is the single interval containing $x = 0$.

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